

example
1)
— 0.5
— 1
— 0.5
on — 1
learning
— 1

Q9: Amureg
a) Bias

formula $\rightarrow \frac{1}{2}$ mark

calc $\rightarrow \frac{1}{2}$ mark

answer $\rightarrow \begin{bmatrix} -7.25 \\ -7.25 \\ 0 \\ 7.25 \end{bmatrix}$ (norm)
OR $\sqrt{141.25 \times 11.88}$
any 1 $\rightarrow$ 1 mark

variance:

Deterministic $\rightarrow$ No variance

Reason $\rightarrow$ 1 mark

OR formula $\rightarrow \frac{1}{2}$
calc $\rightarrow \frac{1}{2}$

1 mark
variance over data points
gets no marks!

1 mark

b)

Accurate model $\rightarrow$ No bias

Reason $\rightarrow$ 1 mark

OR formula $\rightarrow \frac{1}{2}$
calc $\rightarrow \frac{1}{2}$

1 mark

Variance same as (a)

+2 (substitution)
1 mark

(answer)
2 marks

Q10 Roy

a) Step - 8 1 point

Reasoning - Min Val Error
- 1 point

b) No - 0.5 points

Absence of variance / irreducible
error - 0.5

Bias-Variance tradeoff - 1 point

c) 7.7 - 1 point

Explanation - Underfitting / Overfitting
or explaining how val error changes
- 1 point

d) If only overfitting mentioned or low
bias - 0.5 point

Model starts learning noise, and
the noise in the train set may
not be the same as val set
- 1.5 point

Only papers
with query come
main table
for stapling

Q8) - Jai

a) SL - ①
UL - ①

b) SL - 0.5/example
UL - " "

c) SL → Yes - ①
Reason - ①

UL → No - ①
Reason - ①

Other way of learning
- ①

Q9) c) $MSE = Bias^2 + Var + \sigma^2$ } 1 mark

$= 0 + 2$ (substitution)

$= 2$ (answer)

2 marks

Q9: Answer

a) Bias

formula → 1/2 mark

calc → 1/2 mark

answer → $\begin{bmatrix} -7.25 \\ -4.25 \\ 0 \\ 4.25 \\ 7.25 \end{bmatrix}$

variance:

Deterministic → No

Reason → 1 mark

OR formula → 1/2
calc → 1/2

variance
gets

b) Accurate model →

Reason → 1 mark

OR formula → 1/2
calc → 1/2

Variance same a